Austin Huang

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Professional Summary

Robotics and automation systems engineer and technical leader who takes complex robotic systems from prototype to dependable production. Primary technical owner for XtalPi's U.S. automation operations and leader of its U.S. automation engineering team, spanning system architecture, hardware–software integration, perception, production release, reliability, and client delivery. Hands-on across industrial robots, AGV/AMR, computer vision, calibration, simulation, and an AI operations platform that gives engineers, operators, and leads shared operational context through LLM agents.

Professional Experience

XtalPi Inc. Boston, MA

Manager, Automation Systems & Delivery – North America *Mar. 2026 – Present*

- Serve as the primary technical owner for automation across U.S. operations, setting architecture and technical direction across concurrent pharmaceutical programs and acting as the escalation point for system-level decisions.
- Lead XtalPi's U.S. automation engineering team, building it from a one-person function into a team that independently develops, validates, and delivers system updates in the field and in the Boston lab.
- Built the engineering capacity behind the order-of-magnitude growth of XtalPi's North American automation business, from its first deployment to a multi-million-dollar program.
- Own end-to-end technical delivery across North American client programs, including a Fortune 500 pharmaceutical partner: site-readiness engineering, FAT/SAT acceptance evidence, production startup support, and upgrade roadmaps.
- Direct API and workflow releases from lab validation through on-site production promotion, including full-flow validation campaigns that introduced new capabilities with zero regression to legacy workflows.
- Lead the architecture and development of XtalPi's AI operations platform, integrating live robot telemetry, execution logs, and engineering knowledge into shared operational context for LLM agents across automation teams.
- Engineer LLM agents for live troubleshooting, postmortem root-cause analysis, and preflight validation, correlating robot, workflow, and resource state to turn operational evidence into actionable diagnoses, reusable knowledge, and automated checks.
- Resolve technical priorities and deployment tradeoffs across client and engineering teams in English and Chinese, spanning validation strategy, system upgrades, and the technical basis for procurement decisions.

XtalPi Inc. Boston, MA

Automation Engineer *Aug. 2024 – Mar. 2026*

- Joined as the first Automation Engineer in XtalPi's U.S. organization; built, integrated, and deployed a full-scale automated chemistry lab combining ABB robot arms, AGV/AMR, machine vision, motion logic, storage modules, custom hardware, and API workflows.
- Served as primary on-site system owner at a Fortune 500 pharmaceutical partner, raising pass rates on FAT/SAT-defined task evaluations from 59% to 100% under real operating constraints by using ablation testing to isolate failure modes and engineering out sensor drift.
- Cut operational failures by 94% during production stabilization of the fully automated chemistry lab through system-level improvements to task coordination, vision-guided manipulation, calibration, and failure handling.
- Developed a monocular, contour-based 5-DoF end-effector pose-correction method using geometric modeling and least-squares regression; validated randomized pose-error cases in MuJoCo while reducing hardware dependency and deployment complexity.
- Developed an error model that raised the rate of application results landing within jointly agreed target ranges from 29% to 98% across hundreds of validation cases.
- Led customer-facing validation, live demonstrations, troubleshooting, and stabilization, supporting customer adoption of XtalPi's first U.S. commercial platform.

Axle Informatics / NCATS, NIH Rockville, MD

Automation Research Associate (Internship) *Sep. 2023 – Dec. 2023*

- Designed and deployed a High-Density Storage subsystem – structure validated in Femap – integrating Omron AMRs with PF400 robot arms and contributing to a 37% reduction in sample retrieval time.
- Improved vision-based detection accuracy from 53% to 98% by redesigning the image-processing pipeline and implementing shape-detection algorithms in Python and OpenCV.

Laboratory for Computational Sensing and Robotics, Johns Hopkins Baltimore, MD
Research Assistant, Advisor: Russell H. Taylor *May 2023 – Sep. 2023*

- Cut malaria vaccine production setup time by 62% with a computer-vision pipeline designed for a production-like laboratory environment and day-to-day workflow integration.
- Achieved 96% region-of-interest detection accuracy with YOLOv5 and 95% vaccine-quality prediction accuracy; applied PyTorch class-activation mapping to make model decisions interpretable.

Dynacolor Inc. Taipei, Taiwan
R&D Engineer *May 2019 – Mar. 2021*

- Owned mechanical design of critical hardware for a 6-DoF surgical camera system used in fluorescence-guided procedures – spanning optical alignment, environmental protection, and manufacturability.
- Designed components to meet IP66/IP67 ingress-protection and IK10 impact-resistance requirements, taking designs from Creo CAD and GD&T through 2D manufacturing drawings and vendor collaboration across die casting, sheet metal, and injection molding.
- Cut product defects by more than 80% through root-cause analysis, process optimization, and standardized operating procedures.

iQIYI – Clash Bots Beijing, China
Robotics Engineer *Aug. 2017 – Jun. 2018*

- Designed and built a 110 kg combat robot for a robotics competition series that surpassed one billion cumulative views; led a four-person team through an eight-month concept-to-arena build.
- Designed impact-resistant mechanisms in Creo, implemented RF control and motor/contactor systems, and performed welding, wiring, fabrication, testing, and full electromechanical assembly.

Selected Robotics Projects

VRPelviSim – Virtual Simulation for Fluoroscopic Surgery | *Unity, MRTK, DeepDRR, Docker* 2024

- Built a multi-user virtual operating-room environment for surgical training and clinical data collection using Unity, MRTK, CT-derived patient models, and DeepDRR-simulated radiographs.
- Enabled web-based administration through ZeroMQ and Docker; selected as a Best Project Award Finalist at CISST ERC.

Sensor-Based UR5 Manipulation | *ROS, MATLAB, C++, Motion Planning* 2022

- Cut UR5 control time by more than 50%, improved RRT planning efficiency by 73% with a C++ k-d tree, and achieved 0.2 mm end-effector error using Park–Martin hand-eye calibration.

Semantic Segmentation with Transfer Learning | *PyTorch, DeepLabv3+, Python* 2023

- Created a self-collected urban-scene dataset and fine-tuned DeepLabv3+ with a MobileNet backbone, improving pixel accuracy from 66% to 81% and mean IoU from 61% to 78%.

Education

Johns Hopkins University Baltimore, MD
Master of Science in Engineering in Robotics, GPA: 3.9 *Aug. 2022 – May 2024*

National Taiwan University Taipei, Taiwan
Bachelor of Science in Mechanical Engineering *Sep. 2014 – Jun. 2018*

Technical Skills

Robotics & Systems: System architecture and integration, ABB robots, AGV/AMR, ROS/ROS 2, robotic manipulation, motion planning, robot calibration, FAT/SAT, verification and validation, reliability engineering
Perception & Simulation: Computer vision, OpenCV, YOLO, PyTorch, deep learning, camera and hand-eye calibration, MuJoCo, Unity, DeepDRR
Software & AI: Python, C++, MATLAB, C#, Linux, Git, Docker, API integration, LLM agents, agentic workflows, real-time fault diagnostics, logging and knowledge-base pipelines
Mechanical & Delivery: Creo Parametric, GD&T, Femap, design for manufacturing, root-cause analysis, technical leadership, client delivery